# Quantitative Methods for Management

Tutorial Exercise Sheet: Sensitivity and Parametric Analysis

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**Recap: Sensitivity Analysis.** Sensitivity analysis examines whether an optimal basis remains optimal when the data of a linear program change. To perform the sensitivity analysis, we first need to know what the changes influence. To identify this, we can use tools from the simplex algorithm. Denote as $\mathbf{A}$ the constraint matrix including the slack columns. Similarly, $\mathbf{x}$ includes the slack variables, and their coefficients in $\mathbf{c}$ are zero.

Denote as $B$ the indices of the basic variables and $N$ those of the nonbasic variables. The columns indexed by $B$ form a square, invertible matrix $\mathbf{A}_B$; the remaining columns form $\mathbf{A}_N$. Setting $\mathbf{x}_N = 0$ gives the basic solution $\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b}$. The two conditions to check are

$$\begin{aligned} \mathbf{A}_B^{-1}\mathbf{b} \geq 0 \quad &\text{(feasibility)}, \\ \mathbf{r}_N^T = \mathbf{c}_N^T - \mathbf{c}_B^T\mathbf{A}_B^{-1}\mathbf{A}_N \leq 0 \quad &\text{(optimality for maximization)}. \end{aligned}$$

All inequalities hold componentwise. The first condition ensures that the basic variables are nonnegative. The second ensures that increasing a nonbasic variable from zero cannot improve the objective. If both conditions hold, the basis is optimal. For a minimization problem, the reduced-cost condition is instead $\mathbf{r}_N \geq 0$.

The reduced costs also follow from the dual variables (shadow prices):

$$\mathbf{u}^* = (\mathbf{A}_B^T)^{-1}\mathbf{c}_B, \qquad \mathbf{r} = \mathbf{c} - \mathbf{A}^T\mathbf{u}^*, \qquad \mathbf{r}_B = 0.$$

Thus, a reduced cost compares a variable's objective coefficient with the value of the resources it uses. Include the slack variables when checking the reduced-cost condition.

Starting from an optimal basis, consider each of the following changes separately:

- **Only c changes: check optimality.** The basic solution is unchanged and remains feasible. Recompute the reduced costs to determine whether the same basis is still optimal.
- **Only b changes: check feasibility.** The reduced costs are unchanged. Recompute $\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b}$; the basis remains optimal as long as all basic variables remain nonnegative. The production quantities may change even though the basis stays the same.
- **Only $\mathbf{A}_N$ changes: check optimality.** Since the nonbasic variables are zero, the basic solution is unchanged. Recompute the reduced costs using the new nonbasic columns.
- **Coefficients in $\mathbf{A}_B$ change: check both conditions.** First verify that $\mathbf{A}_B$ is still invertible; otherwise, these columns no longer form a basis. Then recompute both the basic solution and the reduced costs.

If a check fails, the original basis no longer satisfies both conditions, and the changed problem requires further analysis.

**Exercise 1** (Sensitivity Analysis)**.** A company produces three types of pasta: spaghetti, penne, and fusilli. Each week, it has 100 hours of shaping machine time and 80 hours of drying equipment time available. Producing 100 kg of spaghetti requires 2 hours of shaping and 1 hour of drying, and generates a net profit of 400 €. Producing 100 kg of penne requires 1 hour of shaping and 2 hours of drying, and generates a net profit of 500 €. Producing 100 kg of fusilli requires 1 hour of shaping and 1 hour of drying, and generates a net profit of 250 €. The company aims to maximize their profit.

(a) Define the decision variables and formulate the company's production problem as a linear program. Proceed to formulate the dual problem and solve it. Determine an optimal production plan and the maximum weekly profit. Interpret the optimal dual variables.

(b) Solve the primal problem again, this time using the simplex algorithm. Compare the result with Part (a).

(c) Suppose the available shaping time increases from 100 to 130 hours per week. Does the original optimal basis remain optimal? Determine the new production plan and weekly profit. Relate the profit increase to the shadow price of shaping capacity.

(d) Suppose the net profit from spaghetti decreases from 400 € to 350 € per 100 kg. Does the original production plan remain optimal? Determine the resulting maximum weekly profit.

(e) Suppose the net profit from fusilli increases from 250 € to 350 € per 100 kg. Use the original dual solution to check whether the original production plan remains optimal. Explain whether the company should now produce fusilli.

Analyze the changes in Parts (c)–(e) separately, starting from the original program.

*Solution.*

(a) **Steps using the dual problem:**

(1) Define the decision variables and formulate the primal objective and constraints.
(2) Formulate the dual, draw its feasible region, and compare the objective values at its vertices to find the minimum.
(3) Use complementary slackness to identify zero primal variables and binding resource constraints; solve for the remaining variables.
(4) Check primal feasibility and equality of the primal and dual objective values. Interpret the production plan, profit, and shadow prices.

Let $x_1$, $x_2$, and $x_3$ denote the weekly quantities of spaghetti, penne, and fusilli, respectively, measured in units of 100 kg. In matrix and vector notation, the profit-maximization problem is

$$\begin{aligned} \max_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b}, \\ & \mathbf{x} \geq 0, \end{aligned}$$

where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} 400 \\ 500 \\ 250 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 100 \\ 80 \end{pmatrix}.$$

The rows of $\mathbf{A}$ represent shaping and drying, and its columns represent spaghetti, penne, and fusilli. All inequalities hold componentwise. The objective $\mathbf{c}^T\mathbf{x}$ is weekly net profit in euros.

Let $\mathbf{u} = (u_1, u_2)^T \geq 0$ contain the dual variables for shaping and drying. The dual then minimizes:

$$\begin{aligned} \min_{\mathbf{u}} \quad & \mathbf{b}^T\mathbf{u} \\ \text{subject to} \quad & \mathbf{A}^T\mathbf{u} \geq \mathbf{c}, \\ & \mathbf{u} \geq 0. \end{aligned}$$

Here,

$$\mathbf{A}^T\mathbf{u} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 2u_1 + u_2 \\ u_1 + 2u_2 \\ u_1 + u_2 \end{pmatrix} \geq \begin{pmatrix} 400 \\ 500 \\ 250 \end{pmatrix}, \qquad \mathbf{b}^T\mathbf{u} = 100u_1 + 80u_2.$$

The feasible region lies above all three constraint lines in the nonnegative quadrant. Its vertices and objective values are

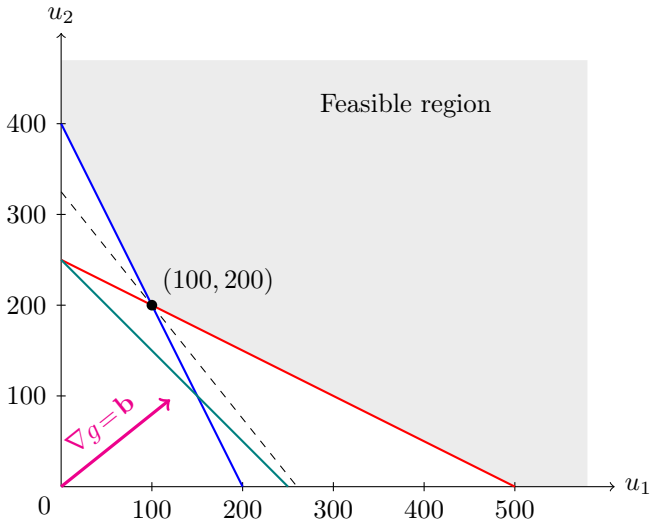
| $(u_1, u_2)$ | $\mathbf{b}^T\mathbf{u}$ |
|---|---|
| $(0, 400)$ | 32,000 |
| $(100, 200)$ | 26,000 |
| $(500, 0)$ | 50,000 |

The minimum occurs at $\mathbf{u}^* = (100, 200)^T$, where the spaghetti and penne constraints intersect.

The graph shows the dual objective $g(\mathbf{u}) = \mathbf{b}^T\mathbf{u} = 100u_1 + 80u_2$. Its gradient is

$$\nabla g(\mathbf{u}) = \begin{pmatrix} \frac{\partial g}{\partial u_1} \\ \frac{\partial g}{\partial u_2} \end{pmatrix} = \begin{pmatrix} 100 \\ 80 \end{pmatrix} = \mathbf{b}.$$

This direction is perpendicular to the objective lines. Since the dual is a minimization problem, we move these lines in the opposite direction until they last touch the feasible region.




Blue: $2u_1 + u_2 = 400$; red: $u_1 + 2u_2 = 500$; teal: $u_1 + u_2 = 250$.
Dashed: $\mathbf{b}^T\mathbf{u} = 26{,}000$; magenta arrow: direction of $\nabla g = \mathbf{b}$.
Both axes use the same scale and are in euros per hour.

Next, we try to find the basis of the optimal solution. For this, we know that the optimality conditions require

$$u_i^*(\mathbf{b} - \mathbf{A}\mathbf{x})_i = 0 \quad (i = 1, 2), \qquad x_j(\mathbf{A}^T\mathbf{u}^* - \mathbf{c})_j = 0 \quad (j = 1, 2, 3).$$

Both dual variables are nonzero, so both resource constraints bind. Also,

$$\mathbf{A}^T\mathbf{u}^* - \mathbf{c} = \begin{pmatrix} 400 \\ 500 \\ 300 \end{pmatrix} - \begin{pmatrix} 400 \\ 500 \\ 250 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 50 \end{pmatrix},$$

so $x_3^* = 0$. Writing $\mathbf{B}$ for the first two columns of $\mathbf{A}$, we obtain

$$\mathbf{B}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{b}, \qquad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{x}^* = \begin{pmatrix} 40 \\ 20 \\ 0 \end{pmatrix}.$$

The primal and dual objective values agree:

$$\mathbf{c}^T\mathbf{x}^* = 400 \cdot 40 + 500 \cdot 20 = 26{,}000 = \mathbf{b}^T\mathbf{u}^*.$$

Thus, the company produces 4,000 kg of spaghetti and 2,000 kg of penne, with no fusilli. Its maximum weekly profit is $\mathbf{c}^T\mathbf{x}^* = 26{,}000$ euros.

The dual variables are profit shadow prices: an additional hour of shaping increases maximum profit by $u_1^* = 100$ euros, while an additional hour of drying increases it by $u_2^* = 200$ euros, as long as the current basis remains optimal.

(b) **Steps using the simplex algorithm:**

(1) Add slack variables and form the initial tableau with the slacks as the basic variables.

(2) Choose the column with the largest negative coefficient in the objective and perform the minimum-ratio test to find the pivot element.

(3) Divide the pivot row by the pivot entry and perform Gauss–Jordan elimination on all other constraint rows. Then, add the pivot row times the cost of the pivot variable to the objective row. Repeat the last two steps until all objective-row coefficients are zero or greater.

(4) Now, read off the optimal basis, their quantities, and the total profit from the tableau.

Introduce the slack vector $\mathbf{s} = (s_1, s_2)^T \geq 0$ for unused shaping and drying time. The constraints become $\mathbf{Ax} + \mathbf{s} = \mathbf{b}$. With $s_1$ and $s_2$ basic, the initial feasible solution is $\mathbf{x} = 0$, $s_1 = 100$, and $s_2 = 80$.

Let $z = \mathbf{c}^T\mathbf{x}$ denote profit. We write the objective row as $z - 400x_1 - 500x_2 - 250x_3 = 0$; the coefficient of $z$ is always one and is omitted from the tableaux. The initial tableau is

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | RHS |
|---|---|---|---|---|---|---|
| $s_1$ | 2 | 1 | 1 | 1 | 0 | 100 |
| $s_2$ | 1 | 2 | 1 | 0 | 1 | 80 |
| $z$ | $-400$ | $-500$ | $-250$ | 0 | 0 | 0 |

A negative objective-row coefficient indicates that increasing the corresponding nonbasic variable can improve profit. Choose $x_2$ to enter because its coefficient, $-500$, is the most negative. The minimum-ratio test gives

$$\min\left\{\frac{100}{1}, \frac{80}{2}\right\} = 40,$$

so $s_2$ leaves. Divide its row by the pivot element 2, then eliminate $x_2$ from the other rows:

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | RHS |
|---|---|---|---|---|---|---|
| $s_1$ | $\frac{3}{2}$ | 0 | $\frac{1}{2}$ | 1 | $-\frac{1}{2}$ | 60 |
| $x_2$ | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 40 |
| $z$ | $-150$ | 0 | 0 | 0 | 250 | 20,000 |

The current solution produces 40 units of penne and earns 20,000 €. The coefficient of $x_1$ is still negative, so $x_1$ enters next. The ratio test is

$$\min\left\{\frac{60}{3/2}, \frac{40}{1/2}\right\} = \min\{40, 80\} = 40,$$

so $s_1$ leaves. Divide its row by $3/2$ and eliminate $x_1$ from the other rows:

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | RHS |
|---|---|---|---|---|---|---|
| $x_1$ | 1 | 0 | $\frac{1}{3}$ | $\frac{2}{3}$ | $-\frac{1}{3}$ | 40 |
| $x_2$ | 0 | 1 | $\frac{1}{3}$ | $-\frac{1}{3}$ | $\frac{2}{3}$ | 20 |
| $z$ | 0 | 0 | 50 | 100 | 200 | 26,000 |

All objective-row coefficients are now nonnegative. In particular, the final objective row gives

$$z = 26{,}000 - 50x_3 - 100s_1 - 200s_2 \leq 26{,}000.$$

Setting the nonbasic variables $x_3 = s_1 = s_2 = 0$ attains this bound. Thus, $\mathbf{x}^* = (40, 20, 0)^T$ is optimal, with weekly profit of 26,000 €, as in Part (a).

To apply the checks in the recap, use the basic variables $(x_1, x_2)$ and the nonbasic variables $(x_3, s_1, s_2)$, in that order. For the constraints including slack variables,

$$\mathbf{A}_B = \mathbf{B}, \qquad \mathbf{A}_B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \mathbf{A}_N = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Both checks hold:

$$\begin{aligned} \mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b} &= \begin{pmatrix} 40 \\ 20 \end{pmatrix} \geq 0, \\ \mathbf{r}_N^T = \mathbf{c}_N^T - \mathbf{c}_B^T\mathbf{A}_B^{-1}\mathbf{A}_N &= \begin{pmatrix} 250 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 100 & 200 \end{pmatrix}\mathbf{A}_N \\ &= \begin{pmatrix} -50 & -100 & -200 \end{pmatrix} \leq 0. \end{aligned}$$

The tableau's objective-row coefficients are the negatives of these reduced costs. In the following parts, $\mathbf{A}$ continues to denote the original constraint matrix; $\mathbf{A}_N$ includes the nonbasic slack columns as above.

(c) Only $\mathbf{b}$ changes, so the reduced costs are unchanged. We only need to check feasibility of the original basis. To compute the inverse of its matrix, first calculate the determinant:

$$\mathbf{A}_B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det(\mathbf{A}_B) = 2 \cdot 2 - 1 \cdot 1 = 3 \neq 0.$$

For a $2 \times 2$ matrix, interchange the diagonal entries, change the signs of the off-diagonal entries, and divide by the determinant. Thus,

$$\mathbf{A}_B^{-1} = \frac{1}{\det(\mathbf{A}_B)}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The feasibility check is then

$$\begin{aligned} \mathbf{x}'_B = \mathbf{A}_B^{-1}\mathbf{b}' &= \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 130 \\ 80 \end{pmatrix} \\ &= \frac{1}{3}\begin{pmatrix} 2 \cdot 130 - 80 \\ -130 + 2 \cdot 80 \end{pmatrix} = \begin{pmatrix} 60 \\ 10 \end{pmatrix} \geq 0. \end{aligned}$$

The condition holds, so the same basis remains optimal. With $x_3 = s_1 = s_2 = 0$, the new plan produces 6,000 kg of spaghetti and 1,000 kg of penne. Weekly profit is $400 \cdot 60 + 500 \cdot 10 = 29{,}000$ euros. The increase of 3,000 € equals the 30 additional shaping hours times the original shadow price of 100 € per hour.

(d) Only $\mathbf{c}$ changes, so the original basic solution remains feasible. We only need to check the reduced costs. Here, $(\mathbf{c}'_B)^T = (350, 500)$ and $\mathbf{c}_N^T = (250, 0, 0)$, with nonbasic variables ordered as $(x_3, s_1, s_2)$. First compute

$$\begin{aligned} (\mathbf{c}'_B)^T\mathbf{A}_B^{-1} &= \frac{1}{3}\begin{pmatrix} 350 & 500 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \\ &= \frac{1}{3}\begin{pmatrix} 2 \cdot 350 - 500 & -350 + 2 \cdot 500 \end{pmatrix} = \begin{pmatrix} 200/3 & 650/3 \end{pmatrix}. \end{aligned}$$

Therefore,

$$\begin{aligned}(\mathbf{r}'_N)^T &= \mathbf{c}_N^T - (\mathbf{c}'_B)^T \mathbf{A}_B^{-1} \mathbf{A}_N \\ &= \begin{pmatrix}250 & 0 & 0\end{pmatrix} - \begin{pmatrix}200/3 & 650/3\end{pmatrix} \begin{pmatrix}1 & 1 & 0 \\ 1 & 0 & 1\end{pmatrix} \\ &= \begin{pmatrix}250 - (200/3 + 650/3) & -200/3 & -650/3\end{pmatrix} \\ &= \begin{pmatrix}-100/3 & -200/3 & -650/3\end{pmatrix} \leq 0.\end{aligned}$$

The condition holds, so the original plan $\mathbf{x}^* = (40, 20, 0)^T$ remains optimal. Weekly profit is $350 \cdot 40 + 500 \cdot 20 = 24{,}000$ euros, a decrease of 2,000 €.

(e) Only the nonbasic objective coefficient $c_3$ changes, so the original basic solution remains feasible and $\mathbf{c}_B$ is unchanged. We only need to check the reduced costs, using the original dual solution $(\mathbf{u}^*)^T = \mathbf{c}_B^T \mathbf{A}_B^{-1} = (100, 200)$:

$$\begin{aligned}(\mathbf{r}'_N)^T &= (\mathbf{c}'_N)^T - (\mathbf{u}^*)^T \mathbf{A}_N \\ &= \begin{pmatrix}350 & 0 & 0\end{pmatrix} - \begin{pmatrix}100 & 200\end{pmatrix} \begin{pmatrix}1 & 1 & 0 \\ 1 & 0 & 1\end{pmatrix} \\ &= \begin{pmatrix}350 - (100 + 200) & -100 & -200\end{pmatrix} \\ &= \begin{pmatrix}50 & -100 & -200\end{pmatrix} \not\leq 0.\end{aligned}$$

The condition fails because the reduced cost of fusilli is $50 > 0$. Both original basic quantities are positive, so fusilli production can increase from zero while adjusting spaghetti and penne production to preserve feasibility. This raises profit by 50 € per additional 100 kg of fusilli while the basic quantities remain nonnegative. Thus, the original plan is no longer optimal, and the company should produce fusilli.

To find the new optimal plan, take the direction $\mathbf{d} = (-1, -1, 3)^T$: reduce spaghetti and penne by one unit each and increase fusilli by three units. Resource use is unchanged, while profit increases:

$$\mathbf{A}\mathbf{d} = \begin{pmatrix}-2 - 1 + 3 \\ -1 - 2 + 3\end{pmatrix} = \begin{pmatrix}0 \\ 0\end{pmatrix}, \qquad (\mathbf{c}')^T \mathbf{d} = -400 - 500 + 3 \cdot 350 = 150.$$

Moving along this direction with step size $t \geq 0$ gives

$$\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d} = \begin{pmatrix}40 - t \\ 20 - t \\ 3t\end{pmatrix} \geq 0 \quad \Longleftrightarrow \quad 0 \leq t \leq 20.$$

Since profit increases with $t$, take $t = 20$. Penne leaves the basis and fusilli enters, giving

$$\mathbf{x}' = \begin{pmatrix}20 \\ 0 \\ 60\end{pmatrix}, \qquad (\mathbf{c}')^T \mathbf{x}' = 400 \cdot 20 + 350 \cdot 60 = 29{,}000.$$

To check optimality of this feasible plan, use the new basic variables $(x_1, x_3)$. Their dual constraints hold with equality:

$$\begin{aligned}2u'_1 + u'_2 &= 400, \\ u'_1 + u'_2 &= 350,\end{aligned} \qquad \Longrightarrow \qquad \mathbf{u}' = \begin{pmatrix}50 \\ 300\end{pmatrix}.$$

For the new nonbasic variables $(x_2, s_1, s_2)$, the reduced costs are

$$\begin{pmatrix} 500 - (50 + 2 \cdot 300) & -50 & -300 \end{pmatrix} = \begin{pmatrix} -150 & -50 & -300 \end{pmatrix} \leq 0.$$

Therefore, the new plan is optimal: produce 2,000 kg of spaghetti and 6,000 kg of fusilli, with no penne, and earn 29,000 € per week.

□

**Recap: Parametric Analysis.** Parametric analysis determines how the optimal solution and objective value depend on a parameter. When only the objective coefficients depend on a parameter $\rho$, the feasible region stays fixed and does not depend on the parameter. If this region is nonempty and bounded, as in the following exercise, we can proceed as follows:

- **Compute the vertices of the feasible region.** In two dimensions, find the intersections of pairs of constraint boundary lines, including the nonnegativity boundaries. Keep only the points that satisfy all constraints. Denote the distinct vertices by $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(k)}$.
- **Compare the objective values at all vertices.** Write the value at vertex $i$ as $z_i(\rho) = \mathbf{c}(\rho)^T \mathbf{x}^{(i)}$. A vertex $\mathbf{x}^{(i)}$ is optimal if and only if

$$\begin{aligned} z_i(\rho) &\geq z_j(\rho) \quad \text{for every } j \quad \text{(maximization)}, \\ z_i(\rho) &\leq z_j(\rho) \quad \text{for every } j \quad \text{(minimization)}. \end{aligned}$$

  For each vertex, solve these inequalities together with the given restrictions on $\rho$ to find the parameter range in which it is optimal.

The optimal objective value is the largest of the functions $z_i(\rho)$ for maximization, or the smallest for minimization. Use the parameter ranges to write it as a piecewise-defined function. At a parameter value where two vertices tie for the optimal value, every point on the line segment joining them is also optimal.

If only the RHS $\mathbf{b} = \mathbf{b}(\rho)$ depends on the parameter, the primal feasible region changes with different values of the parameter $\rho$. However, we can work with the dual:

$$\begin{array}{rl} \max\limits_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \leq \mathbf{b}(\rho), \\ & \mathbf{x} \geq 0, \end{array} \quad \longrightarrow \quad \begin{array}{rl} \min\limits_{\mathbf{u}} & \mathbf{b}(\rho)^T \mathbf{u} \\ \text{subject to} & \mathbf{A}^T \mathbf{u} \geq \mathbf{c}, \\ & \mathbf{u} \geq 0. \end{array}$$

The dual feasible region is fixed because $\mathbf{A}$ and $\mathbf{c}$ do not change with different values of $\rho$. The parameter now appears only in the dual objective coefficients, so we can use the vertex comparisons as above:

- Compute the dual vertices $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(k)}$ and their objective values $w_i(\rho) = \mathbf{b}(\rho)^T \mathbf{u}^{(i)}$.
- Since the dual is a minimization problem, solve $w_i(\rho) \leq w_j(\rho)$ for every $j$ to find the range where vertex $i$ is optimal.

Apply these comparisons for parameter values where the primal is feasible and has a finite optimal value. Strong duality then gives the primal optimal value as $z(\rho) = \min_i w_i(\rho)$. The dual region may be unbounded; a finite minimum is still attained at a vertex for the dual above.

To recover an optimal primal solution, use the optimality conditions with the selected dual optimum $\mathbf{u}^*(\rho)$. If $u_i^*(\rho) > 0$, resource constraint $i$ binds. If $(\mathbf{A}^T\mathbf{u}^*(\rho))_j > c_j$, set $x_j = 0$. Solve the resulting equations and check all primal constraints, including nonnegativity. This gives the optimal quantities as functions of the parameter.

**Exercise 2** (Parametric Analysis of an Objective Coefficient)**.** For the real parameter $\rho \geq 0$, consider the linear programming problem

$$\begin{aligned} \max_{\mathbf{x}} \quad & \rho x_1 + 2x_2 \\ \text{subject to} \quad & 4x_1 + 3x_2 \leq 22, \\ & 2x_1 + x_2 \leq 10, \\ & 2x_1 + 5x_2 \leq 10, \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) Determine the optimal solutions as a function of $\rho$. Write down the problem in vector-matrix notation and check if any constraints are redundant.

(b) Write the optimal objective value $z(\rho)$ as a piecewise-defined function and sketch its graph.

(c) Determine the optimal solution when $\rho = 1$. Find the interval of parameter values for which this solution remains optimal.

*Solution.*

(a) In matrix and vector notation, the problem is $\max\{\mathbf{c}(\rho)^T\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \geq 0\}$, where

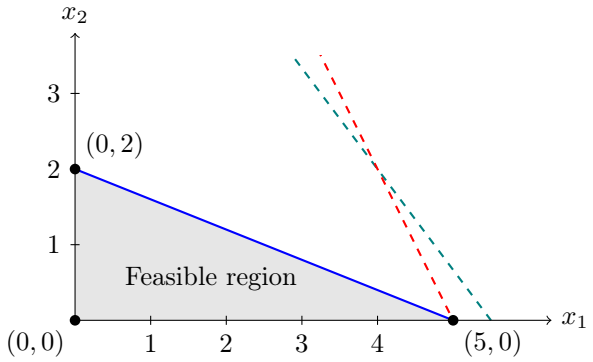
$$\mathbf{c}(\rho) = \begin{pmatrix} \rho \\ 2 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 4 & 3 \\ 2 & 1 \\ 2 & 5 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 22 \\ 10 \\ 10 \end{pmatrix}.$$

Only the payoff vector depends on $\rho$. The first two constraints are redundant: for nonnegative $x_1, x_2$, the third constraint implies

$$\begin{aligned} 4x_1 + 3x_2 &\leq 2(2x_1 + 5x_2) \leq 20 < 22, \\ 2x_1 + x_2 &\leq 2x_1 + 5x_2 \leq 10. \end{aligned}$$

Thus, the feasible region is the triangle with vertices $(0,0)$, $(0,2)$, and $(5,0)$. Their objective values are

| $(x_1, x_2)$ | $\mathbf{c}(\rho)^T\mathbf{x}$ |
|---|---|
| $(0,0)$ | $0$ |
| $(0,2)$ | $4$ |
| $(5,0)$ | $5\rho$ |




Blue: $2x_1 + 5x_2 = 10$; teal dashed: $4x_1 + 3x_2 = 22$;
red dashed: $2x_1 + x_2 = 10$. The shaded triangle is the feasible region.

Checking where the payoffs match, i.e., setting $4 = 5\rho$, gives the breakpoint $\rho = 4/5$. The unique optimal solution away from this breakpoint is

$$\mathbf{x}^*(\rho) = \begin{cases} (0,2)^T, & 0 \le \rho < 4/5, \\ (5,0)^T, & \rho > 4/5. \end{cases}$$

At $\rho = 4/5$, the objective is

$$\frac{4}{5}x_1 + 2x_2 = \frac{2}{5}(2x_1 + 5x_2).$$

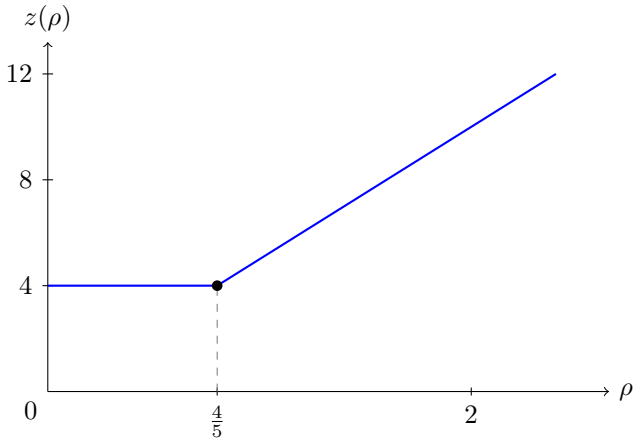
Every point on the edge $2x_1 + 5x_2 = 10$ therefore has objective value 4. All optimal solutions at the breakpoint are of form

$$\mathbf{x} = \begin{pmatrix} s \\ 2 - \frac{2}{5}s \end{pmatrix}, \qquad 0 \le s \le 5.$$

(b) The optimal value is the larger of the two vertex values:

$$z(\rho) = \max\{4, 5\rho\} = \begin{cases} 4, & 0 \le \rho \le 4/5, \\ 5\rho, & \rho > 4/5. \end{cases}$$

The graph is continuous at $\rho = 4/5$. Its slope is zero below the breakpoint and 5 above it.



(c) Since $1 > 4/5$, the unique optimal solution at $\rho = 1$ is $\mathbf{x}^* = (5, 0)^T$, with $z(1) = 5$. This point remains optimal whenever $5\rho \geq 4$, so the required interval is

$$\rho \in \left[\frac{4}{5}, \infty\right).$$

At $\rho = 4/5$, it is one of infinitely many optimal solutions; it is the unique optimal solution for $\rho > 4/5$.

□

**Exercise 3** (Parametric Analysis)**.** A company produces both orange and apple juice. Producing 100 liters of orange juice requires 2 hours of blending and 1 hour of bottling, and generates a net profit of 40 €. Producing 100 liters of apple juice requires 1 hour of blending and 1 hour of bottling, and generates a net profit of 30 €. The company has 16 hours of blending time and 10 hours of bottling time available per day. The company aims to maximize their profit.

(a) Define the decision variables and formulate the company's production problem as a linear programming problem. Solve the problem graphically. Determine the optimal production plan, maximum daily profit, and binding constraints.

For the following exercises you may use the solution of Part (a):

$$\mathbf{c} = \begin{pmatrix} 40 \\ 30 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 16 \\ 10 \end{pmatrix}, \quad \mathbf{x}^* = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

(b) Suppose the available blending time is now $t$ hours per day, where $t \geq 0$, while bottling time remains fixed at 10 hours. Formulate and solve the dual problem as a function of $t$. Use complementary slackness to determine the optimal production plan and identify the parameter values at which the binding constraints change.

(c) Use the dual solution to write the maximum daily profit $V(t)$ as a piecewise-defined function and sketch its graph. Interpret the slope on each interval using shadow prices.

(d) Use the shadow prices to determine the smallest blending capacity beyond which additional blending time does not increase profit. Explain which resource limits production in this range.

(e) Using the optimal shadow-price vector $\mathbf{u}^*(t)$ from Part (b), compute the reduced cost vector $\mathbf{r}(t) = \mathbf{c} - \mathbf{A}^T\mathbf{u}^*(t)$. Use it together with complementary slackness to recover the optimal production plan in each parameter range without applying the simplex algorithm.

*Solution.*

(a) Let $x_1$ and $x_2$ denote the daily quantities of orange juice and apple juice, respectively, measured in units of 100 liters. The problem is

$$\begin{aligned} \max_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b}, \\ & \mathbf{x} \geq 0, \end{aligned}$$

where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} 40 \\ 30 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 16 \\ 10 \end{pmatrix}.$$

The rows of $\mathbf{A}$ represent blending and bottling, and its columns represent orange juice and apple juice. The objective $\mathbf{c}^T\mathbf{x} = 40x_1 + 30x_2$ is daily net profit in euros.

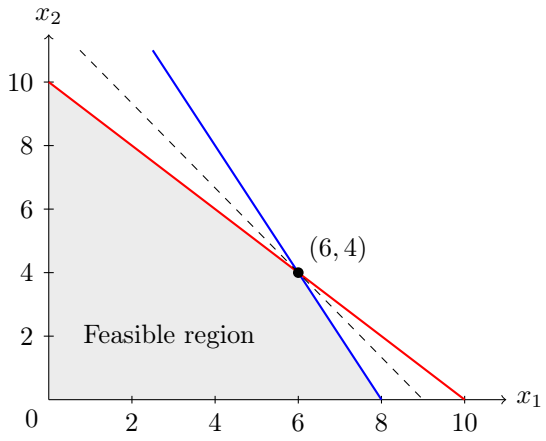
The feasible region lies below $2x_1 + x_2 = 16$ and $x_1 + x_2 = 10$ in the nonnegative quadrant. The two lines intersect at

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \qquad \Longrightarrow \qquad \mathbf{x} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}.$$

Evaluating profit at the vertices gives

| $(x_1, x_2)$ | $\mathbf{c}^T\mathbf{x}$ |
|---|---|
| $(0, 0)$ | 0 |
| $(8, 0)$ | 320 |
| $(6, 4)$ | 360 |
| $(0, 10)$ | 300 |

Thus, $\mathbf{x}^* = (6, 4)^T$ is optimal. The company produces 600 liters of orange juice and 400 liters of apple juice, earning 360 € per day. Both resource constraints bind, and neither nonnegativity constraint binds.




Blue: $2x_1 + x_2 = 16$; red: $x_1 + x_2 = 10$.
Dashed: $40x_1 + 30x_2 = 360$. Quantities are in units of 100 liters.

(b) The resource vector is $\mathbf{b}(t) = (t, 10)^T$. Let $\mathbf{u} = (u_1, u_2)^T \geq 0$ denote the shadow prices of blending and bottling. The dual is

$$\begin{aligned} \min_{\mathbf{u}} \quad & \mathbf{b}(t)^T\mathbf{u} = tu_1 + 10u_2 \\ \text{subject to} \quad & \mathbf{A}^T\mathbf{u} \geq \mathbf{c}, \\ & \mathbf{u} \geq 0, \end{aligned} \qquad \mathbf{A}^T\mathbf{u} = \begin{pmatrix} 2u_1 + u_2 \\ u_1 + u_2 \end{pmatrix} \geq \begin{pmatrix} 40 \\ 30 \end{pmatrix}.$$

Its feasible region does not depend on $t$. The vertices and their objective values are

| $(u_1, u_2)$ | $\mathbf{b}(t)^T\mathbf{u}$ |
|---|---|
| $(30, 0)$ | $30t$ |
| $(10, 20)$ | $10t + 200$ |
| $(0, 40)$ | 400 |

Comparing these values gives $30t \leq 10t+200$ exactly when $t \leq 10$, and $10t+200 \leq 400$ exactly when $t \leq 20$. Thus, one optimal dual solution for every $t \geq 0$ is

$$\mathbf{u}^*(t) = \begin{cases} (30,0)^T, & 0 \leq t \leq 10, \\ (10,20)^T, & 10 < t < 20, \\ (0,40)^T, & t \geq 20. \end{cases}$$

At $t = 10$ and $t = 20$, adjacent dual vertices tie, so the dual optimum is not unique.

**How to use complementary slackness.** For feasible primal and dual solutions, optimality holds if each shadow price times the unused amount of its resource is zero, and each production quantity times the slack in its dual constraint is zero. Thus, a positive shadow price forces the corresponding resource constraint to bind. A strict dual inequality forces the corresponding production quantity to be zero. If a shadow price or a dual slack is zero, its product condition gives no further restriction; we use the remaining equations and check feasibility.

Here, the four complementary slackness equations are

$$u_1^*(t - 2x_1 - x_2) = 0, \quad u_2^*(10 - x_1 - x_2) = 0,$$
$$x_1(2u_1^* + u_2^* - 40) = 0, \quad x_2(u_1^* + u_2^* - 30) = 0.$$

We substitute the selected dual optimum in each parameter range:

- **For $0 \leq t \leq 10$, use $\mathbf{u}^* = (30,0)^T$.** The orange-juice dual slack is $2 \cdot 30 + 0 - 40 = 20$, so $20x_1 = 0$ gives $x_1 = 0$. Since $u_1^* = 30 > 0$, blending binds:

  $$30(t - 2x_1 - x_2) = 0 \quad \Longrightarrow \quad x_2 = t.$$

  This plan is feasible because $x_2 = t \geq 0$ and bottling use is $x_1 + x_2 = t \leq 10$. The apple-juice dual slack is $30 + 0 - 30 = 0$, so its product condition is also satisfied.

- **For $10 < t < 20$, use $\mathbf{u}^* = (10,20)^T$.** Both shadow prices are positive, so

  $$10(t - 2x_1 - x_2) = 0, \qquad 20(10 - x_1 - x_2) = 0.$$

  Hence, $2x_1 + x_2 = t$ and $x_1 + x_2 = 10$. Subtracting the second equation from the first gives $x_1 = t - 10$, and substitution gives $x_2 = 20 - t$. Equivalently,

  $$\mathbf{x}^*(t) = \mathbf{A}^{-1}\mathbf{b}(t) = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} t \\ 10 \end{pmatrix} = \begin{pmatrix} t - 10 \\ 20 - t \end{pmatrix}.$$

  Both quantities are positive in this range, and both resource constraints hold with equality. The dual slacks are $2 \cdot 10 + 20 - 40 = 0$ and $10 + 20 - 30 = 0$, so the remaining product conditions hold.

- **For $t \geq 20$, use $\mathbf{u}^* = (0,40)^T$.** The apple-juice dual slack is $0 + 40 - 30 = 10$, so $10x_2 = 0$ gives $x_2 = 0$. Since $u_2^* = 40 > 0$, bottling binds:

  $$40(10 - x_1 - x_2) = 0 \quad \Longrightarrow \quad x_1 = 10.$$

  Blending use is $2x_1 + x_2 = 20 \leq t$, so the plan is feasible. The orange-juice dual slack is $2 \cdot 0 + 40 - 40 = 0$, so its product condition is also satisfied.

Therefore,

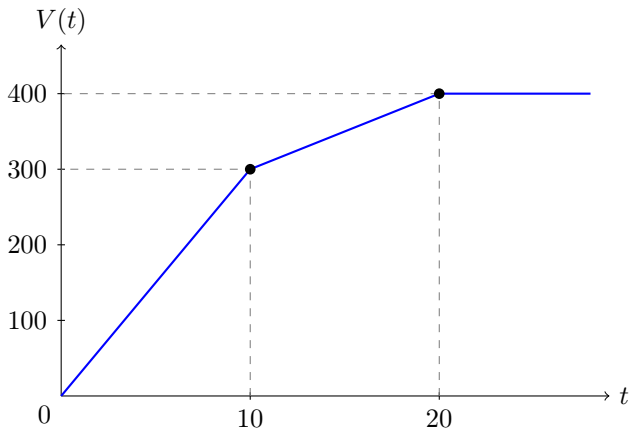
$$\mathbf{x}^*(t) = \begin{cases} (0,t)^T, & 0 \le t \le 10, \\ (t-10, 20-t)^T, & 10 < t < 20, \\ (10,0)^T, & t \ge 20. \end{cases}$$

Each plan is feasible and satisfies $\mathbf{c}^T\mathbf{x}^*(t) = \mathbf{b}(t)^T\mathbf{u}^*(t)$, confirming optimality. For $0 < t < 10$, only blending binds among the resource constraints, and $x_1 = 0$. For $10 < t < 20$, both resource constraints bind and both products are produced. For $t > 20$, only bottling binds, and $x_2 = 0$. At $t = 10$ and $t = 20$, both resource constraints bind and one production quantity is zero. At $t = 0$, only the zero production plan is feasible.

(c) By strong duality, the maximum profit equals the minimum dual objective:

$$V(t) = \mathbf{b}(t)^T\mathbf{u}^*(t) = \min\{30t, 10t + 200, 400\} = \begin{cases} 30t, & 0 \le t \le 10, \\ 10t + 200, & 10 < t < 20, \\ 400, & t \ge 20. \end{cases}$$

The pieces agree at the breakpoints: $V(10) = 300$ and $V(20) = 400$. For the original capacity, $\mathbf{u}^*(16) = (10, 20)^T$ gives $V(16) = 16 \cdot 10 + 10 \cdot 20 = 360$, as in Part (a).




Blending capacity $t$ is measured in hours per day; $V(t)$ is daily profit in euros.

Within each open parameter interval, the dual solution is constant and

$$V'(t) = u_1^*(t) = \begin{cases} 30, & 0 < t < 10, \\ 10, & 10 < t < 20, \\ 0, & t > 20. \end{cases}$$

Thus, an additional blending hour raises profit by 30 €, 10 €, or zero, respectively, provided the change stays within the same interval. At the breakpoints, the dual optimum is not unique and $V$ has no single derivative. For a small capacity increase, use the slope to the right: 10 at $t = 10$ and zero at $t = 20$.

(d) For $t \ge 20$, the dual solution $\mathbf{u}^* = (0, 40)^T$ gives

$$V(t) = \mathbf{b}(t)^T\mathbf{u}^* = t \cdot 0 + 10 \cdot 40 = 400.$$

Blending has zero value for further capacity increases, while bottling has a positive shadow price. Since $V(t) < 400$ for $t < 20$, the smallest blending capacity beyond which additional time has no value is 20 hours per day. The company produces 1,000 liters of orange juice and earns 400 € per day. For $t > 20$, unused blending time is $t - 20$ hours, while all 10 bottling hours remain in use. At $t = 20$, both constraints bind, but increasing blending capacity alone still gives no improvement.

(e) The reduced cost vector measures each product's unit profit minus the value of the resources it uses at the optimal shadow prices:

$$\mathbf{r}(t) = \mathbf{c} - \mathbf{A}^T\mathbf{u}^*(t) = \begin{pmatrix} 40 - (2u_1^* + u_2^*) \\ 30 - (u_1^* + u_2^*) \end{pmatrix}.$$

This speeds up the calculation because the reduced cost vector contains the negatives of all dual slacks. The selected dual solution is constant on each parameter range, so we compute this vector only once per range. Its negative entries immediately identify products with zero production. Together with the positive shadow prices, this leaves only the binding resource equations to solve, without writing all complementary slackness equations again.

Dual feasibility gives $\mathbf{r}(t) \leq 0$ and the optimality conditions require

$$x_j^* r_j(t) = 0 \quad (j = 1, 2), \qquad u_i^*(\mathbf{b}(t) - \mathbf{A}\mathbf{x}^*)_i = 0 \quad (i = 1, 2).$$

Therefore, a negative reduced cost forces the corresponding production quantity to zero, while a positive shadow price forces the corresponding resource constraint to bind. Using the dual solutions selected in Part (b) gives

| Range | $\mathbf{u}^*(t)$ | $\mathbf{r}(t)$ | Conditions on $\mathbf{x}^*$ |
|---|---|---|---|
| $0 \leq t \leq 10$ | $(30, 0)^T$ | $(-20, 0)^T$ | $x_1 = 0, \quad 2x_1 + x_2 = t$ |
| $10 < t < 20$ | $(10, 20)^T$ | $(0, 0)^T$ | $\mathbf{A}\mathbf{x} = \mathbf{b}(t)$ |
| $t \geq 20$ | $(0, 40)^T$ | $(0, -10)^T$ | $x_2 = 0, \quad x_1 + x_2 = 10$ |

Solving these simple equations immediately gives $(0, t)^T$, $(t - 10, 20 - t)^T$, and $(10, 0)^T$, respectively. A zero reduced cost does not by itself imply positive production; for example, $\mathbf{r} = (0, 0)^T$ in the middle range, and the two positive shadow prices determine the quantities through the binding resource constraints.

□